# RFI Response: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

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Public feedback will inform HHS-wide use of **three different approaches**: regulation, reimbursement, and research & development. In general, HHS seeks feedback on ways in which these approaches can be most effectively applied to **support the rapid adoption and use of AI in clinical care**, to foster public trust and confidence in modern technology solutions, to reduce uncertainty that impedes AI innovation, and to align federal incentives so that AI is deployed in ways that enhance productivity, reduce burden, lower health care costs, and improve health outcomes for patients, caregivers, and communities.

Hypothetically, if a Payer is taking financial risk for the long-term health and health costs of an individual, that payer will have an inherent incentive to promote access to the **highest-value interventions for patients**. We seek feedback on payment policy changes that ensure payers have the incentive and ability to promote **access to high-**

**value AI clinical interventions**, foster competition among clinical care AI tool builders, and accelerate access to and affordability of AI tools for clinical care.

## Specific Questions

In addition to the general requests for information above regarding AI regulation, reimbursement, and research & development, HHS seeks input on the following specific questions:

### 1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

The private sector's ability to innovate in AI for healthcare is hindered by a complex interplay of regulatory, technical, economic, and human factors. These barriers not only slow the development of new AI tools but also impede their adoption and integration into clinical care.

1. **Regulatory and Compliance Hurdles**

   Stringent regulations, while essential for patient safety, create lengthy approval processes that delay innovation. For instance, the FDA approval timeline for AI models often spans 12-18 months, which is seen as excessively slow compared to private sector norms. This regulatory uncertainty stifles rapid iteration and deployment, with stakeholders calling for clearer frameworks to balance innovation with risk management.

2. **Data Access, Privacy, and Quality Issues**

   AI relies on vast, high-quality datasets, but healthcare data is often siloed, heterogeneous, and protected by privacy regulations, making it difficult for private innovators to train robust models. Over one-third of life sciences professionals report data accuracy, security, and quality as major obstacles, with issues like biased or incomplete datasets leading to unreliable AI outputs. In clinical settings, limited access to anonymized patient data hinders model evaluation and refinement, contributing to products that don't meet clinician expectations. A 2025/26 report highlights that 52% of organizations view data quality and availability as primary barriers to AI adoption across sectors, including healthcare.

   Empowered-Home addresses these issues with a HIPAA compliant architecture that deidentifies the patient from the PHI when querying the LLM for correlations and insights and reassembles the response in a secure framework. Privacy follows patient individual right of access allowing them to consent to limited view sharing of their vital trends with non-licensed caregivers. They can also consent to have their compendium displayed in the clinician dashboard to be shared with specialists as well as the sponsoring Primary Care Physician.

   Quality issues of LLM hallucination are addressed be controlling the source information that the LLM can query to peer reviewed medical journals or other trusted sources. In addition, we use BERT that dramatically improves accuracy for large language models. It is an architecture that incorporates Bidirectional Encoder Representations from Transformers (BERT) that learns to represent text as a sequence of vectors using self-supervised learning.  It uses the encoder-only transformer architecture that the LLM can understand. Finally, we have the ability to use consensus-based reasoning whereby we can incorporate more than one LLM connection.  If two disagree, a third model can be queried to find consensus in truth.

Sources:

- https://www.weforum.org/stories/2026/01/digital-solutions-and-ai-in-healthcare
- https://intuitionlabs.ai/articles/ai-adoption-life-sciences-barriers
- https://downloads.regulations.gov/HHS-ONC-2026-0001-0031/attachment_1.pdf

### 3. Reimbursement and Economic Challenges

Misaligned financial incentives, where payment systems reward treatment over prevention, discourage investment in preventive AI tools. High development costs and unclear return on investment (ROI) affect 51% of organizations, despite 42% reporting strong returns from AI. Legacy reimbursement models fail to cover many AI interventions, limiting affordability and scalability. Funding shortages and capital allocation trade-offs further impede private sector progress, as organizations struggle to justify AI investments amid competing priorities.

Sources:

- https://downloads.regulations.gov/HHS-ONC-2026-0001-0045/attachment_2.pdf
- https://pratikmistry.medium.com/the-main-barriers-slowing-ai-adoption-in-healthcare-in-2026-based-on-research-36552cbb9ea9
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- https://www.hinshawlaw.com/en/insights/healthcare-alert/hhs-seeks-input-on-adoption-of-ai-in-clinical-care-and-strengthening-supply-chain-resilience
- https://www.forbes.com/sites/johnbremen/2026/02/17/overcoming-barriers-to-ai-adoption-in-2026/

### D. Integration and Infrastructure Limitations

Outdated legacy IT systems in healthcare facilities present significant obstacles to integrating AI into workflows which is why Empowered-Home uses a web-based app for clinicians that will work in any secure browser like Chrome.  Challenges include poor interoperability, data fragmentation, and the need for substantial infrastructure upgrades, which are acute in rural or under-resourced settings where IT connectivity lags by up to 64% compared to urban areas.  This results in many AI tools that are difficult to deploy at scale, with 67% of implementations in U.S. hospitals misaligned with actual needs like addressing provider shortages.

Empowered-Home is architected in a way that makes sharing patients' vitals trends, clinical risk alerts, and correlative insights, ubiquitous.  We use CDS Hooks to integrate with existing EMR workflows and connect to the EMR systems through FHIR.  We use an Interoperability as a Service platform to handle file format conversion to and from FHIR.  We also use the same platform to connect to the health record ecosystem through QHINs (Commonwell Health Alliance, CareQuality, and eHealth Exchange) combined with a Record Locator

Service (RLS) to discover patient health records scattered across the ecosystem. This enriches the patient record of the sponsoring physician adding value the continuous data we collect on patients.

Sources:

- https://pratikmistry.medium.com/the-main-barriers-slowing-ai-adoption-in-healthcare-in-2026-based-on-research-36552cbb9ea9
- https://www.weforum.org/stories/2026/01/digital-solutions-and-ai-in-healthcare

### E. Workforce Resistance and Skills Gaps

Clinicians often resist AI due to concerns over safety, role changes, and lack of trust, compounded by burnout and time constraints. Organizational skill deficits in AI—both technical and soft skills—are cited by many leaders as a leading barrier. Change management issues, including inadequate training and governance, further slow adoption.

At Empowered-Home, we took these issues into account when designing our modern Remote Health Monitoring as a Service platform. Because the interface is so intuitive, unfamiliar users can start using it without problems, immediately. The graphic design is such that it fosters trust because vitals trends can be easily viewed and drilled into. Problem areas are identified with color coding that match with the AI insights and correlations. Physicians can assign roles and responsibilities to their clinician team members with access controls and audit logs of who made what changes to vital thresholds (for example). The platform can be used to enhance the training and ability for clinicians to conduct effective triage with the AI supporting them along the way. Urban clinicians can augment rural teams enhancing clinician support and training, allowing clinicians to work at the top of their license.

## 2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

## Key Priorities for HHS to Incentivize Effective AI Use in Clinical Care

### A. Regulatory Changes

Establish Proportionate-to-Risk Oversight and Expedited Approval Pathways. HHS should prioritize creating flexible regulatory frameworks under the FDA and ONC that differentiate between high-risk (e.g., autonomous decision-making AI) and low-risk applications (e.g., supportive tools for risk stratification or clinical decision support – Empowered-Home's model), including expedited reviews for validated AI tools. This includes issuing guidance on human-in-the-loop requirements, bias assessments, and post-deployment monitoring to clarify accountability without prescriptive mandates.

**Why?** Current ambiguity and lengthy approvals stifle private sector innovation, deter partnerships, and delay patient access to tools that could improve diagnostic accuracy and reduce errors during triage. Proportionate regulation builds trust, prevents privacy breaches (e.g., HIPAA 45 CFR Parts 160/164). It also avoids repeating EHR mandate pitfalls where heavy-handed rules favored compliance over outcomes. That lead to billions in spending without health improvements.

**B. Enhance Data Interoperability and Privacy Standards**:

Update regulations to mandate standardized, secure data sharing via encrypted infrastructure in a FHIR framework to reduce the expense load on data transformation! All EMR systems should be <u>required</u> to produce and accept FHIR compatible document formats. No more CCDAs in XML or Discharge Summaries in HL7v2! <u>Those are OLD data formats and standards</u>. It should also include patient-mediated access to complete records while strengthening HIPAA protections against data misuse in AI training. That's what Empowered-Home does.

**Why?** Fragmented data silos across 400+ EHR systems hinder AI model training and real-world performance, exacerbating biases and limiting adoption in diverse populations. Clear standards would enable organic innovation, protect patient privacy, and create liquid datasets for better personalization, ultimately lowering costs and improving equity. Empowered-Home is architected to store the records in the patient's own, personal Apple Health account. Not all vitals and health data from devices maps 1:1 into Apple Health, so we do the LOINC code mapping and ensure its stored neatly there for the patient to take with them.

**C. Payment Policy Changes**

**Shift to Outcomes-Oriented, Value-Based Reimbursement Models:** HHS, through CMS, should modernize legacy fee-for-service systems to reward AI-enabled activities that demonstrate cost reductions or improved outcomes, such as preventive care, chronic disease management, and reduced hospitalizations. This includes creating temporary Medicare codes for AI tools, incentivizing payers to cover high-value interventions, and avoiding separate billing for AI to prevent inflation.

**Why?** Traditional models incentivize treatment over prevention, externalizing savings and discouraging AI investments despite proven benefits like lower ED visits and enhanced efficiency (e.g., 40-50% of physicians already use AI organically, but not necessarily safely). Reforms would align incentives for affordability, foster competition among developers, and accelerate access, potentially reducing healthcare's 18% GDP share while improving patient experiences.

**D. Incentivize Adoption Through Augmented Reimbursement and Pilots**:

Introduce financial bonuses for providers using AI in quality workflows such as Medicare Advantage or ACOs; and penalties for non-adoption, while ensuring inclusion of diverse stakeholders like Certified Registered Nurse Anesthetists (CRNAs) in policy development.

**Why?** Without incentives, short-term costs deter uptake, especially in under-resourced settings; targeted rewards would de-risk early adoption, promote equity, and generate evidence for scaling, addressing barriers like administrative burden and ensuring AI supports rather than replaces clinicians.

**E. Develop Standardized Evaluation Frameworks and Transparency Measures:**

Create industry-aligned tools for assessing AI validity, workflow integration, and patient-facing disclosures, with HHS publishing top needs and preferred product lists to guide adoption.

**Why?** Fragmented evaluations cause duplication and delays. Standardized, transparent processes would ease procurement, enhance trust (e.g., plain-language disclosures), and mitigate risks like hallucinations or biases, ensuring AI prioritizes patient safety and equity.

## HHS Regulations, Policies, and Programs to Revisit for Enhancing AI Capabilities in Clinical Care

### 1. HIPAA Privacy and Security Rules (45 CFR Parts 160 and 164)

Specific Changes: Update guidance to explicitly address AI-specific scenarios, such as using de-identified or synthetic PHI for model training and inference. Clarify that sharing anonymized datasets for AI development in secure environments (e.g., data trusts or federated learning) is permissible under business associate agreements, with strengthened audit requirements. Make the HIPAA Security Rule requirements voluntary for low-risk AI applications to reduce compliance burdens, while mandating risk assessments for AI handling PHI. Additionally, revise the Breach Notification Rule (45 CFR § 164.404) to exempt or streamline reporting for breaches affecting fewer than 500 individuals if they involve AI-generated insights rather than raw PHI.

Rationale: Current ambiguities hinder AI development by limiting access to diverse datasets needed for robust models, potentially biasing outputs to companies with fewer financial resources slowing innovation. Clearer rules would allow AI to process aggregated data for clinical insights (e.g., predicting disease risks) while protecting privacy, fostering trust and accelerating adoption. This aligns with calls to prevent PHI disclosure to general-purpose AI without governance.

**2. Confidentiality of Substance Use Disorder Records (42 CFR Part 2)**

Specific Changes: Align 42 CFR Part 2 with HIPAA standards by removing restrictions on sharing substance use disorder (SUD) records for AI-driven care coordination and research, so long as patient consent is obtained and data is de-identified wherever possible. Revise § 2.11 to expand definitions of "qualified service organizations" to include AI developers under strict privacy safeguards, and update consent requirements (§ 2.31) to allow electronic or AI-assisted consents for data use in predictive analytics.

Rationale: The current patchwork limits comprehensive datasets for AI training, impeding tools for holistic care (e.g., integrating SUD data with mental health AI models). Harmonization would enable AIs to provide more accurate, integrated clinical support, reducing barriers for hospitals and improving outcomes in behavioral health without new mandates.

**3. ONC Health IT Certification Program (45 CFR Part 170)**

Specific Changes: In § 170.315(b)(11), remove or simplify the "model card" transparency requirements for AI-driven Decision Support Interventions (DSIs), limiting them to high-risk applications only. Reserve or eliminate expired criteria like § 170.315(a)(9) for clinical decision support (CDS), and revise § 170.315(a)(5) to drop non-essential patient observation data elements (§ 170.315(a)(5)(i)(D)-(H)) unless tied to specific program measures. Promote voluntary certification for AI interoperability, incorporating standards like FHIR for AI data exchange.

Rationale: Overly prescriptive criteria burden developers and limit flexible AI integration into workflows. Deregulation would allow AIs to evolve rapidly (e.g., real-time CDS without redundant documentation), reducing costs and enabling broader use in certified health IT. An example of this is the HTI-5 Proposed Rule's focus on prosperity through reduced oversight.

**4. Common Rule for Human Subjects Research (45 CFR Part 46)**

Specific Changes: Reexamine the definition of "identifiable private information" under § 46.102(e)(7) to account for AI's ability to re-identify data through dataset combinations, requiring periodic reviews every two years and mandating advanced de-identification techniques like differential privacy for AI research datasets.

Rationale: Evolving AI capabilities challenge outdated identifiability concepts, restricting research data for model training. Updates would facilitate ethical AI development such as using anonymized EHR data for diagnostic tools, fulfilling the Common Rule's commitment to adapt to technology while protecting subjects.

**5. FDA Medical Device Regulations (21 CFR Parts 801, 803, 807, 820, and 830)**

Specific Changes: Clarify guidance for AI/ML software as a medical device (SaMD) by creating expedited pathways under for low-risk AI updates such as continuous learning models like Empowered-Home without full premarket review, emphasizing post-market surveillance (§ 803). Revise Quality System Regulations (§ 820) to include AI-specific risk management, such as bias audits, while exempting non-diagnostic AI tools from device classification if they support human review.

Rationale: Ambiguous pathways slow AI deployment in clinical settings. Streamlined processes would enable AIs to assist in diagnostics or monitoring more readily, balancing innovation with safety as per FDA's evolving framework.

In summary, these revisions would create a more enabling environment for AI in clinical care by prioritizing risk-proportionate oversight and data liquidity.  HHS should collaborate with stakeholders via pilots or sandboxes to test changes, ensuring they enhance desired capabilities—such as generating evidence-based insights—while maintaining equity and trust.

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (*e.g.,* relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

**Novel Legal and Implementation Issues for Non-Medical Device AI in Clinical Care**

- Liability uncertainty:  Who is responsible for AI errors—developer, clinician, or hospital? It's unclear who bears liability for harms—clinicians who use it, developers who design it, hospitals who buy it, or AI vendors create it.  HHS should provide clearer frameworks and safe harbors to promote innovation during this critical time.
- Indemnification gaps:  Non-standard contracts and uneven risk distribution.  HHS should provide best-practice guidance and Task Force expansion.
- Privacy challenges:  Re-identification risks, consent gaps, outdated HIPAA for AI training/ HHS should provide us with targeted de-identification rules and secure data-use standards.

- Security vulnerabilities: Model attacks, amplified breach risks.  Non-device AI often lacks mandatory FDA cybersecurity guidance.  HHS can help by providing us with updated breach notification requirements and safety monitoring.
- Bias/discrimination:  Inequitable outcomes from skewed data. HHS can help us with funded audits, transparency mandates, and civil rights enforcement.
- Broader trust/implementation friction: Opacity, workforce resistance, regulatory fragmentation.  HHS can help by improving explainability, literacy/training, and harmonized rules across stakeholders.

## Recommended Role for HHS in Addressing These Issues

HHS should issue targeted guidance and standards—such as AI-specific HIPAA updates for de-identification and breach notifications, plus clear liability frameworks with safe harbors for vetted tools—to reduce uncertainty around privacy, security, and responsibility.

Additionally, HHS can promote accountability to cover liability and indemnification best practices.  Liability insurance is very expensive and hurts real innovation coming from startups. HHS could help fund pilots, mandating explainability certification, fostering state-industry collaboration and workforce training grants.  It's also advisable that HHS establishes an AI Safety Program to monitor incidents and enforce equity and nondiscrimination.

HHS involvement as a supporter of AI innovation would directly enable more companies specializing in AI-powered remote health monitoring solutions like Empowered-Home to move quickly and gain higher adoption that meets the goals of HHS.  They can achieve this by reducing regulatory uncertainty and privacy/security barriers for their AI-driven data from devices, clarifying responsibility in multi-party care scenarios (e.g., clinicians vs. developers for monitoring errors), facilitating easier EHR integration through required FHIR interoperability between EMR systems, and bias mitigation in diverse populations like rural or maternal groups. They can also help tremendously by providing funding for pilots aligned with rural health initiatives, and aligning reimbursement models (e.g., CMS value-based care) to incentivize adoption -- ultimately allowing faster innovation, scaling, and broader deployment.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (*e.g.,* contracts, grants, cooperative agreements, and/or prize competitions)?

## Promising AI Evaluation Methods for Non-Medical Devices in Clinical Care

Non-medical device AI—tools like predictive analytics, workflow optimization, or decision support software (not classified as medical devices)—should require evaluation to ensure safety, equity, and effectiveness in clinical settings.  The most promising methods emphasize iterative, risk-proportionate approaches that integrate technical validation with real-world human-centered evaluation. These align with principles from organizations like the American Heart Association (AHA) and address gaps in traditional metrics by accounting for expert variability and contextual shifts.  Below, are outlined key methods, categorized by phase and type.

**Pre-deployment methods**

- Validation on diverse datasets representing rural, maternal, senior, and underserved populations.
- Robustness testing against real-world variations such as inconsistent wearable data from inferior quality devices (we've done a lot of testing), connectivity issues in rural areas, and environmental noise.
- Human-centered evaluations like usability testing, cognitive load assessment, pilot programs that can scale from small to large population sets.

**Post-deployment methods** focus on continuous monitoring:

- Performance tracking from discharge to stable state, care plan management and tracking to improve outcomes of health.
- Real-world outcome linkage (e.g., readmissions, time-to-decision, clinician satisfaction).
- Accountability metrics and understanding of patient population improvements.

HHS should further support these processes to bridge gaps in infrastructure, standardization, and evidence for non-regulated AI, accelerating safe adoption while reducing uncertainty for developers and providers.

**Grants to fund real-world pilots**, longitudinal monitoring frameworks, robustness studies in diverse home settings, and human-centered evaluation tools.

Contracts for HHS-facilitated validation environments, standardized reporting templates, or vendor-neutral testing platforms that could validate Empowered-Home's non-device AI in real clinical workflows.

These mechanisms would build shared infrastructure, generate credible evidence, impact on ED reduction and cost savings, reduce adoption uncertainty for providers, and align with HHS goals for trustworthy AI in remote and home-based clinical care.

### 5. How can HHS best support private sector activities (*e.g.,* accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

HHS can best support private sector activities in accreditation, certification, industry-driven testing, and credentialing for AI in clinical care by acting as a convener, validator, and **funder** —fostering alignment without stifling innovation.

By leveraging these supports, HHS can reduce fragmentation, boost public trust, and drive equitable AI adoption—directly empowering innovators like Empowered-Home to extend clinical care into homes, aligning with the HHS AI Strategy for better outcomes and affordability.

### 6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

AI tools deployed in clinical care have shown mixed results: notable successes in narrow, high-value applications contrasted with frequent shortfalls in broader integration and sustained impact.

**Where AI Has Met or Exceeded Expectations**

- **Ambient AI Scribes and Documentation Tools**: These have significantly reduced clinician "pajama time" (after-hours charting) by 40-60% in many deployments (e.g., Kaiser Permanente's large-scale rollout of Abridge across 40+ hospitals, Nuance DAX Copilot, and Microsoft Dragon Copilot at Advocate Health). They improve workflow efficiency, allow more patient-focused time, and deliver clear ROI through time savings without harming care quality.
- **Radiology and Imaging AI**: Tools for detection, prioritization, and triage (e.g., Aidoc, Rad AI, Google DeepMind for breast cancer) often match or exceed human accuracy (e.g., 94% vs. 65% in lung nodule detection), reducing false negatives/positives, speeding reads, and supporting better outcomes with fewer missed cases.
- **Predictive Analytics and Early Detection**: In areas like sepsis detection, cardiovascular risk prediction, or antibiotic-resistant infections, AI has improved accuracy, reduced readmissions/hospitalizations, and cut costs (e.g., preventing 200+ readmissions in some studies, saving millions annually). Broader estimates suggest potential system-wide savings of $200-360 billion/year through optimized diagnostics, fraud detection, and administrative automation.

**Where AI Has Fallen Short**

- **Broader Clinical Decision Support and Workflow Integration**: Many tools fail to scale due to legacy EHR fragmentation, poor interoperability, clinician resistance, silent underuse, or lack of reimbursement pathways. Diagnostic reasoning AI often stalls, with hallucinations or context loss leading to errors. Up to 80% of initiatives fail from misalignment with billing/compliance systems, not technical flaws.
- **Administrative and Operational Overhype**: While scribes succeed, other applications such as full predictive models or claims denial tools face trust issues, hidden costs, or quiet failures where clinicians work around them, absorbing time savings into fuller schedules rather than reinvesting in care.
- **Equity and Real-World Variability**: Biases persist in underrepresented data, and tools underperform in diverse or rural settings without robust validation.

### Novel AI Tools with Greatest Potential to Improve Outcomes, Quality Insights, and Reduce Costs

Tools shifting toward preventive, home-based, and proactive care show strong promise, especially for chronic conditions, aging populations, and value-based models:

- **AI-Powered Remote Patient Monitoring (RPM) with Generative AI Summarization**: Continuous insights from wearables/in-home devices detect subtle changes early, enabling proactive interventions, reducing ED visits/hospitalizations, and personalizing care.
- **Predictive Risk Stratification and Virtual Companions**: For high-risk groups (e.g., seniors, maternal health, rural patients), these provide triage, emotional support, and coordinated alerts, improving adherence, outcomes, and cost efficiency.
- **Hybrid Clinical Decision Support**: Evidence-based GenAI for triage, personalized plans, and population health targeting high-risk cohorts.

AI in telemedicine is transforming rural healthcare by addressing longstanding barriers like geographic isolation, specialist shortages, limited infrastructure, and workforce constraints. As of early 2026, AI integration enhances remote consultations, diagnostics, monitoring, and triage, making care more accessible, efficient, and proactive—especially for chronic conditions, maternal health, seniors, and preventive services.

### Key Benefits and Applications

- Improved Access and Early Detection: AI-powered tools analyze patient data (e.g., symptoms, imaging, wearables) during virtual visits to provide faster, more accurate diagnostics and triage. This reduces unnecessary travel and enables early interventions for issues like diabetic retinopathy, stroke, or chronic diseases—critical in areas with few specialists.

- Remote Patient Monitoring (RPM) and Predictive Analytics: AI processes real-time data from in-home devices or wearables to detect subtle changes, predict risks (e.g., exacerbations), and alert providers. This supports continuous care without frequent in-person visits, lowering hospitalizations and ED use.
- Decision Support and Second Opinions: AI offers real-time guidance to rural clinicians (e.g., ambient documentation, pattern recognition in records, or virtual specialist-like insights), reducing diagnostic errors and burnout while extending expertise across distances.
- Hybrid and Mobile Models: Emerging approaches include AI-assisted mobile clinics (e.g., electric vehicles with integrated devices for lab procedures and diagnostics) and chatbots/virtual assistants for initial assessments, overcoming broadband limitations in some cases via low-bandwidth tools.
- Administrative Efficiency: AI streamlines scheduling, claims, and follow-ups, freeing rural providers for patient care.

Studies and reviews (e.g., systematic literature from 2025) highlight that combining AI with telemedicine bridges gaps, improves screening/early detection, and enhances equity, with potential for better long-term outcomes in underserved populations.

**Real-World Examples and Progress**

- AI-Enhanced Platforms: Tools like AI-driven second opinions or diagnostic aids help rural providers access specialist-level insights quickly, as seen in deployments for teledermatology or telestroke.
- Mobile AI Solutions: Projects like University of Florida's Multi-Tags (AI-guided lab procedures in electric vehicles) and University of Michigan's AI-powered mobile clinics enable hospital-level care in remote areas.
- Broader Trends: Rural hospitals adopt AI for virtual care extensions, revenue cycle support, and hybrid models, with 2025-2026 trends emphasizing EHR integration, specialized telemedicine, and cybersecurity.

Challenges persist, including data biases, transparency concerns, infrastructure (e.g., broadband gaps), ethical issues (e.g., unbiased algorithms), and implementation costs/barriers in low-resource settings. Adoption often starts small (e.g., ambient AI for documentation) to build trust and demonstrate value.

**Connection to Empowered-Home.com**

Empowered-Home.com stands out as a practical example tailored to rural needs. The platform uses AI-powered monitoring that integrates consumer/medical wearables and in-home devices for continuous insights, generating clinician summaries and predictive risk alerts. It focuses on rural/remote patients, maternal health, seniors/assisted living, and chronic condition monitoring—extending care boundaries, enabling proactive interventions, reducing unnecessary ED visits/hospitalizations, and supporting evidence-based, patient-centered decisions. By combining IoT, generative AI, and clinical expertise, it aligns with high-impact telemedicine trends, promoting wellness in areas with limited access while addressing cost and outcome goals in value-based care models.

Overall, AI in rural telemedicine is accelerating in 2025-2026, with strong evidence of benefits in access and efficiency. Equitable scaling requires addressing infrastructure, privacy, and validation hurdles through partnerships and policy support.

## 7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

The roles, decision makers, or governing bodies with the most influence on AI adoption for clinical care in healthcare organizations primarily include the following:

- C-suite executives (e.g., CEOs, CIOs, CMIOs, and Chief Strategy/Clinical Officers): They control budgets, set strategic priorities, and drive centralized decisions on AI investments and use cases, often outpacing general IT spending.
- Clinical leaders and department heads (e.g., physicians, chief medical officers, and specialty leads): They influence adoption through workflow integration, trust-building, and championing tools that fit clinical practice.
- IT and informatics leaders (e.g., CIOs, data science teams): They handle technical feasibility, integration with EHRs, and infrastructure.
- Multidisciplinary AI governance committees (increasingly common): These include compliance, IT, clinical operations, ethics, legal, and sometimes patient representatives; they oversee evaluation, risk management, transparency, and post-deployment monitoring.
- Healthcare managers/leaders and boards: They shape organizational culture, provide support for training/change management, and align AI with strategy.
- Frontline clinicians (physicians and nurses) exert significant bottom-up influence via acceptance, overrides, and feedback, but top-down leadership often determines scale. Hospitals with professional managers (e.g., integrated salary models) tend to adopt faster than physician-led ones.

**Primary administrative hurdles to AI adoption in clinical care include:**

- Data privacy, security, and regulatory compliance (e.g., HIPAA uncertainties, bias risks, lack of clear governance): Cited by ~60% of leaders as top concerns, delaying implementation.

- Integration and interoperability issues with legacy EHRs and fragmented systems: Hinders seamless workflow embedding.
- Lack of resources/infrastructure (e.g., skilled personnel shortages, poor IT readiness, high costs): Limits evaluation, training, and scaling.
- Workflow misalignment and clinician mistrust/resistance (e.g., black-box concerns, fear of de-skilling, over-reliance risks): Leads to underuse or abandonment despite technical promise.
- Unclear reimbursement and ROI (e.g., no direct payment for many AI tools): Disincentivizes investment amid misaligned incentives.
- Governance and evaluation gaps (e.g., fragmented monitoring, liability concerns): Slows trust and sustained use.

These hurdles are often organizational/sociotechnical rather than purely technical, requiring leadership buy-in, training, and standardized frameworks for progress.

## 8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Enhanced interoperability would most dramatically widen **market opportunities**, fuel **research**, and accelerate **AI development** for clinical care in the following high-impact areas, by enabling seamless, standardized access to richer, longitudinal datasets across fragmented systems.

**Primary Areas of Greatest Impact**

- **Comprehensive Longitudinal Patient Records**: Aggregating EHR data (clinical notes, medications, vitals, allergies), labs, imaging, genomics, social determinants, and claims into unified views. This powers AI for predictive modeling (e.g., readmission risk, personalized treatment), chronic disease management, and population health analytics—unlocking massive markets in value-based care, preventive AI, and precision medicine.
- **Real-World Data for AI Training and Validation**: Combining structured/unstructured clinical data with RPM/wearables providing 24/7 continuous monitoring, telemedicine feeds, and observational datasets. This fuels research on real-world outcomes, bias detection, and generalizable models, while enabling scalable AI tools for early detection, clinical decision support, and drug repurposing.
- **Specialized Clinical Domains**:
    - <u>Oncology/precision medicine</u>: Shared genomic + treatment + outcomes data for AI-driven therapies.

- Behavioral/mental health: Longitudinal records with SDOH for predictive interventions.
- Rural/remote and maternal health: 24/7 continuous RPM data augmented with AI-triage supported telemedicine calls and proactive AI alerts based on clinical severity to help clinicians prioritize patients.

- **Administrative and Operational Data Flows**: Claims, prior auths, scheduling, and billing data for AI in revenue cycle optimization, fraud detection, and workflow efficiency—creating new efficiency-focused markets.

**Specific Data Standards Enabling This**

- **HL7 FHIR**: Core for real-time, API-driven exchange; supports modular resources (e.g., Observation, DiagnosticReport, SBAR, Patient) and direct EHR querying by AI apps. FHIR's flexibility accelerates LLM integration and app development.
- **USCDI (United States Core Data for Interoperability)**: Mandated baseline (via ONC/HTI rules) for allergies, meds, labs, notes, imaging metadata—ensures consistent, nationwide access and reduces vendor-specific silos.
- **OMOP Common Data Model**: Standardizes observational/research data for large-scale AI studies; supports NLP on unstructured text and federated analysis.
- **Supporting terminologies:** SNOMED CT, LOINC, ICD-10 for semantic alignment.

**Benchmarking Tools and Platforms**

- **MedAgentBench / Virtual EHR Sims**: Evaluate AI agents on FHIR APIs in realistic workflows, testing interoperability, reasoning, and performance.
- **qDAR (NIST)**: Measures data quality critical for AI reliability and bias assessment.
- **BigDataBench / Open Leaderboards**: Compare model robustness across interoperable datasets, promoting transparent benchmarking.

Companies like **Zen Healthcare IT** exemplify how enhanced interoperability drives progress. Zen provides the **Gemini Integration-as-a-Service Platform** (with tools like Stargate IHE gateway, eHealth Exchange/Carequality connectivity) to simplify data exchange for digital health vendors, HIEs, ACOs, payers, and providers. By enabling faster, secure connections to national networks and supporting custom adapters/APIs, Zen reduces integration barriers—directly widening market access for AI innovators, fueling research via better data liquidity, and accelerating clinical AI deployment (as emphasized by Zen's president in discussions on AI's dependence on interoperable data for improving outcomes and patient experience).

Prioritizing FHIR/USCDI expansion, TEFCA enforcement, and funded benchmarking (e.g., grants, prizes) would create a more liquid data ecosystem, lowering costs for AI developers and enabling equitable, high-impact tools across care settings.

## 9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

### Challenges Patients and Caregivers Wish AI to Address in Clinical Care

Patients and caregivers see AI as a tool to tackle key pain points in healthcare, based on recent surveys and case studies from 2025-2026:

- **Proactive Support and Early Intervention**: Providing clinical risk alerts based on severity to help them focus on patients with the most urgent need, pattern recognition for symptoms, and context-aware assistance in triage for high-risk groups (e.g., seniors, chronic patients) to prevent escalations and improve outcomes.
- **Empowerment in Decision-Making**: Offering correlations to underlying causes of symptoms to distill root causes, second opinions, and personalized insights to foster equity, accessibility, and human-centered care.
- **Improved Understanding of Medical Information**: Translating complex records, diagnoses, and test results into plain language to reduce confusion and enable faster, informed decisions. Drug-to-drug, food-to-drug, and supplement-to-drug interactions, treatment complications.
- **Streamlined Communication and Preparation**: Summarizing patient histories for providers, preparation for visits to cut delays and boost confidence in care.
- **Enhanced Access and Efficiency**: Supporting chronic disease management, wellness, and administrative tasks like scheduling prioritization, or patient reminders for care plan adherence, especially for caregivers handling routines and relationships. Ability to share 24/7 patient health trends with multiple caregivers to foster collaboration.

### Concerns Patients and Caregivers Have Regarding AI Adoption

While optimistic, patients and caregivers express balanced worries, often prioritizing safeguards:

- **Privacy and Data Security**: Fears of breaches, misuse of personal health data, and re-identification risks in AI systems.
- **Bias, Inaccuracy, and Reliability**: Concerns about algorithmic biases (e.g., racial/ethnic disparities), hallucinations, misdiagnoses, or poor generalization to diverse cases.

- **Loss of Human Touch and Empathy**: Worry that AI dehumanizes care, reduces clinician-patient interactions, and erodes trust or emotional support.
- **Transparency and Accountability**: Lack of disclosure on AI use, unclear responsibility for errors, and potential over-reliance leading to ignored human advice.
- **Equity and Ethical Issues**: Fears of increased costs, access disparities, or misuse in sensitive areas like end-of-life decisions.

## Case Studies:

### Predictive Analytics for Sepsis Detection and Early Intervention

AI enables hours-earlier alerts, saving lives in time-sensitive conditions.

- **Johns Hopkins** (COMPOSER model) implemented an AI system that predicts sepsis before obvious symptoms, reducing mortality by ~17-20% in studies involving hundreds of thousands of patients. It caught deteriorations early, preventing escalations and demonstrating measurable survival gains.
- Similar tools in other systems (e.g., predictive models in general wards) have cut serious adverse events by 35% and cardiac arrests by >86%.
- **Hackensack Meridian** integrated AI with Epic for medication instruction mapping, eliminating manual entry in most cases and improving patient safety/outcomes by addressing unstructured Rx data challenges.
- **Mayo Clinic** invests >$1 billion in 200+ AI projects spanning diagnostics, patient care, and research—extending beyond admin to clinical improvements.
- **Geisinger** uses AI/predictive analytics for care coordination, early disease detection, and resource optimization.

### AI in Telemedicine and Remote Monitoring for Rural/Underserved Access (NEW)

AI bridges gaps in chronic care and remote areas.

- **Empowered-Home.com** uses AI-powered remote patient monitoring integrating wearables and in-home devices for continuous insights, generating clinician summaries and predictive alerts. Focused on rural/remote patients, maternal health, seniors, and chronic conditions, it detects subtle changes early, enables proactive interventions, reduces unnecessary ED visits/hospitalizations, and supports evidence-based decisions—directly tackling access barriers and preventive care in value-based models.

## 10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

a. Are there published findings about the impact of adopted AI tools and their use clinical care?

b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

### Specific Areas of AI Research HHS Should Prioritize to Accelerate Adoption in Clinical Care

The highest impact area HHS should prioritize is research that emphasizes continuous remote patient health monitoring (RPM) with AI, focusing on proactive clinician notifications, correlative insights for symptom triage, and root cause analysis to enable early interventions, reduce hospitalizations, and enhance personalized care in clinical settings. Key areas include:

- **AI-Driven Predictive Analytics for Proactive Notifications**: Develop algorithms that analyze real-time RPM data (e.g., from wearables and in-home devices) to detect subtle deviations, generate automated alerts for clinicians, and support multivariate correlations to identify emerging risks before symptoms escalate.  Old RPM solutions that rely on nurses to monitor patient health data don't scale.  AI needs to be doing that.
- **Correlative Insights and Automated Triage Systems**: Advance AI for synthesizing multimodal data (e.g., vitals, activity, environmental factors) to provide root cause analysis, prioritize high-risk cases, and assist in differential diagnosis during triage, reducing clinician workload and improving accuracy in chronic conditions like heart failure or allergies.
- **Privacy-Preserving Federated Learning in RPM**: Enable secure, decentralized model training on distributed patient data to enhance RPM without centralizing sensitive information, addressing HIPAA while improving model robustness for diverse populations.
- **Human-AI Integration for RPM Workflows**: Research hybrid systems where AI augments clinicians in real-time monitoring and triage, offering explainable insights and notifications to foster trust, reduce burnout, and ensure seamless integration into telehealth and virtual care models.
- **Bias Mitigation and Equity in Continuous Monitoring**: Focus on diverse datasets for RPM AI to minimize disparities in alerts and insights, particularly for rural, elderly, or chronic care patients, with validation in real-world settings.
- **Interoperability and Data Standards for RPM**: Enhance FHIR/USCDI frameworks to support seamless RPM data flows, enabling correlative AI analysis across devices and EHRs for proactive care.
- **Post-Deployment Monitoring and Safety in RPM**: Create benchmarks for drift detection in continuous data streams, ensuring reliable notifications and triage over time.

These priorities, supported by grants and pilots, align with HHS's AI Strategy to shift from reactive to proactive care, particularly in RPM for better outcomes and efficiency.

### A. Published Findings on the Impact of Adopted AI Tools in Clinical Care

Yes, 2025-2026 literature highlights positive impacts in RPM-focused AI, with tools enabling proactive notifications and correlative insights that reduce readmissions and improve triage, though deployment challenges like data integration persist.

- **RPM with Proactive Alerts**: Some AI platforms detect abnormal patterns in RPM data, triggering timely interventions and reducing administrative burden; studies show 46% more detections and 10x fewer false positives in systems like Cleveland Clinic's Bayesian Health rollout.
- **Correlative Insights for Triage**: AI in wearables (e.g., for allergy or cardiac monitoring) provides real-time predictive management, enabling root cause analysis and personalized alerts that cut hospitalizations in heart failure patients.
- **Broader RPM Impacts**: Tools like those in Cityhälsan Centrum's telemedicine program track home parameters for proactive care, improving engagement and outcomes; overall, AI RPM reduces clinician workload by prioritizing risks and empowers patients with insights.
- **Diagnostic and Monitoring Advances**: AI-aided systems (e.g., Johns Hopkins' COMPOSER) predict deteriorations 6-7 hours early, reducing mortality by 17-20%, with similar gains in continuous monitoring for chronic conditions.
- **Adoption Trends**: 22% of organizations use domain-specific RPM AI (7x increase from 2024); failures often from poor integration, but successes show >50% time savings in monitoring workflows.

### B. How the Literature Approaches Costs, Benefits, and Transfers of Using AI in Clinical Care

Literature uses cost-effectiveness analysis (CEA), ROI models, and value frameworks, with a growing emphasis on RPM's proactive elements—quantifying savings from early notifications and triage insights while considering transfers like shifted burdens from hospitals to home-based care.

- **Costs**: Assessed as upfront (e.g., device integration, AI training) and ongoing (e.g., data processing for continuous monitoring), with RPM AI reducing long-term expenses through fewer false alerts; studies highlight needs for bias mitigation to avoid inequitable costs.
- **Benefits**: Measured in reduced readmissions (e.g., $200-360B U.S. savings annually), improved QALYs from proactive interventions, and efficiency gains (e.g., 1.7B clinician hours freed via automated triage and root cause insights in RPM).

- **Transfers**: Evaluated as reallocations (e.g., cost shifts to payers via preventive RPM, value flows to patients through home-based monitoring); frameworks advocate novel reimbursements to capture equity gains in proactive care.

## About Empowered-Home

Empowered-Home is a new and innovative AI-powered health technology platform designed to support healthcare organizations & their clinicians in remotely monitoring their patients' health vitals and supporting health data (sleep, exercise, stress, activity), continuously. It is not a rework of an old-style RPM platform from the Covid days. It was built on a modern and flexible event-driven data orchestration platform that can handle thousands of transactions per second and safely interface with multiple LLMs for consensus reasoning. It also has the ability to safely manage agents performing tasks that clinicians don't have time for, but with a policy administration layer for safety. It was built to quickly adapt to technology improvements and to be flexibly configured for wide variety of programs.

Through the use of modern wearables and in-home devices, the EH platform can detect changes in patients' health stability and auto-generate clinical risk alerts based on severity, helping clinicians prioritize care of their patients while also helping them triage patient symptoms.

Unified data from multiple devices, in combination with a lightweight import of the patient's medical information (discharge summary, encounters, SBAR, and health problems from their electronic health records), is presented to the clinician in a dashboard.

The AI identifies trends, generates correlations, suggests triage questions for the clinician to ask the patient during telemedicine calls.

It provides a framework to monitor the patient through each stage (from discharge to recovery to stability).

A corresponding iPhone app keeps the patient and family members informed of their daily health trends relative to their behavior and environment, allowing caregivers to coordinate care around the patient.

The patient leverages new data sharing privileges under the 21st Century Cures Act and the Individual Right of Access, empowering them to make decisions about who sees what data.

**Who Empowered-Home Benefits:**

- It benefits **hospitals** with reduced Emergency Department visits, reduction in 30-day readmissions, DRG denials, and Length of Stay.
- It benefits **patients** living in remote areas and family members who worry about their health and well-being.

- It benefits **health insurers** who are desperately trying to contain escalating healthcare costs with predictive and preventative care.
- It removes a large workload from **physicians** and allows nurses and community paramedics to operate at the top of their license.
- It also helps solve the **healthcare worker** staff shortage and serves as a training tool for consistency and accuracy.

We hope to have the opportunity to play a significant role in the advancement of HHS's objectives as AI reshapes the efficiency of healthcare.